Jana Integration with Bank Systems

A detailed but non-technical overview for a banking audience. Jana Earth Data, May 2026.

1. The split of responsibilities

The bank already has the data it owns. The loan amount, the borrower identifier, the borrower's enterprise value or revenue, the disbursement date, the internal sector code, the branch, the relationship manager. All of that sits in the bank's core banking system or loan origination system today. Nothing changes about that.

Jana brings the data the bank does not have and would otherwise spend analyst hours collecting per loan. That includes satellite-verified facility emissions from Climate TRACE, national emissions context from EDGAR, plant-level metadata from the Global Cement and Concrete Tracker, ownership trees from the Global Energy Monitor entity registry, and air-quality readings from OpenAQ where station coverage exists.

The bank's borrower is the join key. Jana matches on legal entity name, sector, and physical coordinates. For a Nepali commercial book, that means roughly seventy borrowers (the largest cement, hydropower, manufacturing, hotel, logistics, and waste operators) match directly to facility-level data. Another fifty or so match through capacity-derived estimates from registries like the GCCT. The remainder use sector-benchmark emissions, and retail loans stay out of scope.

2. Three integration patterns

Pick whatever the bank's IT comfort allows. The simplest is batch file exchange, which is what most Nepali banks would default to. The bank exports its commercial loan book and borrower master to a flat file on a weekly or monthly cadence. Jana ingests it, enriches it with emissions, taxonomy classification, and PCAF scoring, and returns the enriched file. The bank loads it back into its data warehouse or directly into the loan file. No new system runs alongside the core banking platform. This is how most of the regulated reporting in Nepali banking already works.

The middle option is a REST API integration at the loan origination point. When a credit officer pulls up a new application in the bank's loan origination system, the system makes a single call to Jana with the borrower's identifier or coordinates, and Jana returns the screening data in real time. The credit officer sees it inline, the loan file captures it, and the data is stamped at the decision moment. This is faster but requires the bank's loan origination team to add the integration to their workflow, which is roughly a month of bank-side IT work.

The third is an embedded widget or single-sign-on portal where the bank's compliance and ESG team uses Jana's dashboard directly, with the bank's own borrower data loaded in. This works well for a Green Unit or sustainable finance team that is already a separate workflow from the front-office credit officers.

3. The operational rhythm

Three time horizons to think about, mapped to the three NRB frameworks.

At loan origination, an ESRM screening fires for every new commercial loan application. Today this is a manual research project per borrower. With Jana the same screening becomes a single lookup that returns the borrower's facility emissions, air-quality context, ownership chain, and a risk classification. The credit officer reviews it, attaches conditions, and submits to the credit committee. The full data trail is captured in the loan file for the bank's quarterly NRB submission.

Quarterly, the bank's full loan book is run through the taxonomy classifier. Green, amber, red, and unclassified shares are computed across the portfolio. NRB requires this submission every quarter and uses it as a supervisory input. Jana's classification engine handles the bulk of this and flags any borrowers that have moved between categories since the prior quarter.

Annually, the NSRS disclosure is assembled. Financed emissions are aggregated across the year, attribution math is computed against each loan's outstanding balance and enterprise value, and a PCAF-compliant disclosure block is produced. This is the figure that goes into the bank's annual sustainability report.

4. What the bank actually needs to provide

In simplest form, six data points per commercial borrower:

(1) Borrower legal name. (2) Borrower registration number or LEI. (3) Physical address or coordinates of the operating facility. (4) NRB sector code. (5) The bank's internal enterprise value or revenue figure. (6) Current loan outstanding.

That is enough to run the full integration. Banks already have all six in their existing loan management system. The integration work is plumbing, not data collection.

5. Phased rollout

Most banks would adopt this in three steps over six to twelve months.

Phase one is a batch enrichment of the existing commercial book to establish the baseline and identify which borrowers match facility data. The Green Unit or sustainable banking team can start using the dashboard immediately on this enriched dataset.

Phase two is the ongoing monthly refresh plus the ESRM lookup at loan origination. This wires the data into the credit workflow.

Phase three is the annual NSRS disclosure pipeline. By this point the bank has a year of standing data and the disclosure is largely automated.

6. The honest bit on cost and effort

The bank-side integration is light. There is no rebuild of the core banking system. The bank's loan origination workflow gets one new field or one new lookup. The reporting team gets a quarterly file exchange and an annual pipeline.

The heavy lift is the policy work, the governance approvals, and the staff training on the regulations, which the bank would be doing for ESRM and the Green Finance Taxonomy anyway. Jana removes the manual data-collection bottleneck that makes those policies hard to operate.

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